



TEMASEK JUNIOR COLLEGE
Preliminary Examinations
Higher 2

CANDIDATE
NAME

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CIVICS GROUP: _____/10

INDEX
NUMBER:

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BIOLOGY

9648/03

Applications Paper and Planning Question

16th September 2011

Paper 3

2 hours

Additional Materials: Paper

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
DO not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
1	
2	
3	
4	
5	
Total	

This document consists of **14** printed pages.

[Turn over

- 1 Table 1.1 summarises examples of genetically modified (GM) crops currently or previously produced.

Crop (main use)	Properties of the GM variety	Modification	Percentage Modified in US	Percentage Modified Worldwide
Soybeans (cattle, swine and chicken feed)	Resistant to herbicides	Herbicide resistant gene taken from bacteria inserted into soybean	93	77
Cotton (textile)	Pest-resistant	Bt crystal protein gene transferred into plant genome	93	49
Hawaiian papaya (fruit)	Resistant to the papaya ringspot virus	New gene added into plant genome	80	-
Tomatoes (vegetable)	Fruit softening after harvesting retarded	Antisense gene of enzyme responsible for softening added into plant genome	Taken off the market due to commercial failure	Small quantities grown in China
Sugar beet (vegetable)	Resistant to herbicides	New genes added into plant genome	Planting under controlled conditions	9

Table 1.1

- (a) Briefly outline the steps to modify soybean, to make it resistant to herbicides.

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.....[3]

- (b) The first commercially grown GM crop was the tomato and it was welcomed by consumers who paid a substantial premium over the price of regular tomatoes.

Suggest what could have then caused the “commercial failure” of this crop.

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.....[2]

- (c) With reference to Table 1.1, explain if GM soybeans or GM sugar beet appears to be more successful.

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.....[2]

In 1907, scientists found that a common plant tumour was caused by the invasion of a bacterium called *Agrobacterium tumefaciens*. They discovered that the tumour was caused when plasmids from the bacteria were taken into the DNA of the host plant's cells and expressed. By 1983, biotechnologists were able to introduce genes into plant cells using modified plasmids which do not result in tumour formation. Fig. 1.1 shows the procedure of transforming *Brassica* using this method.

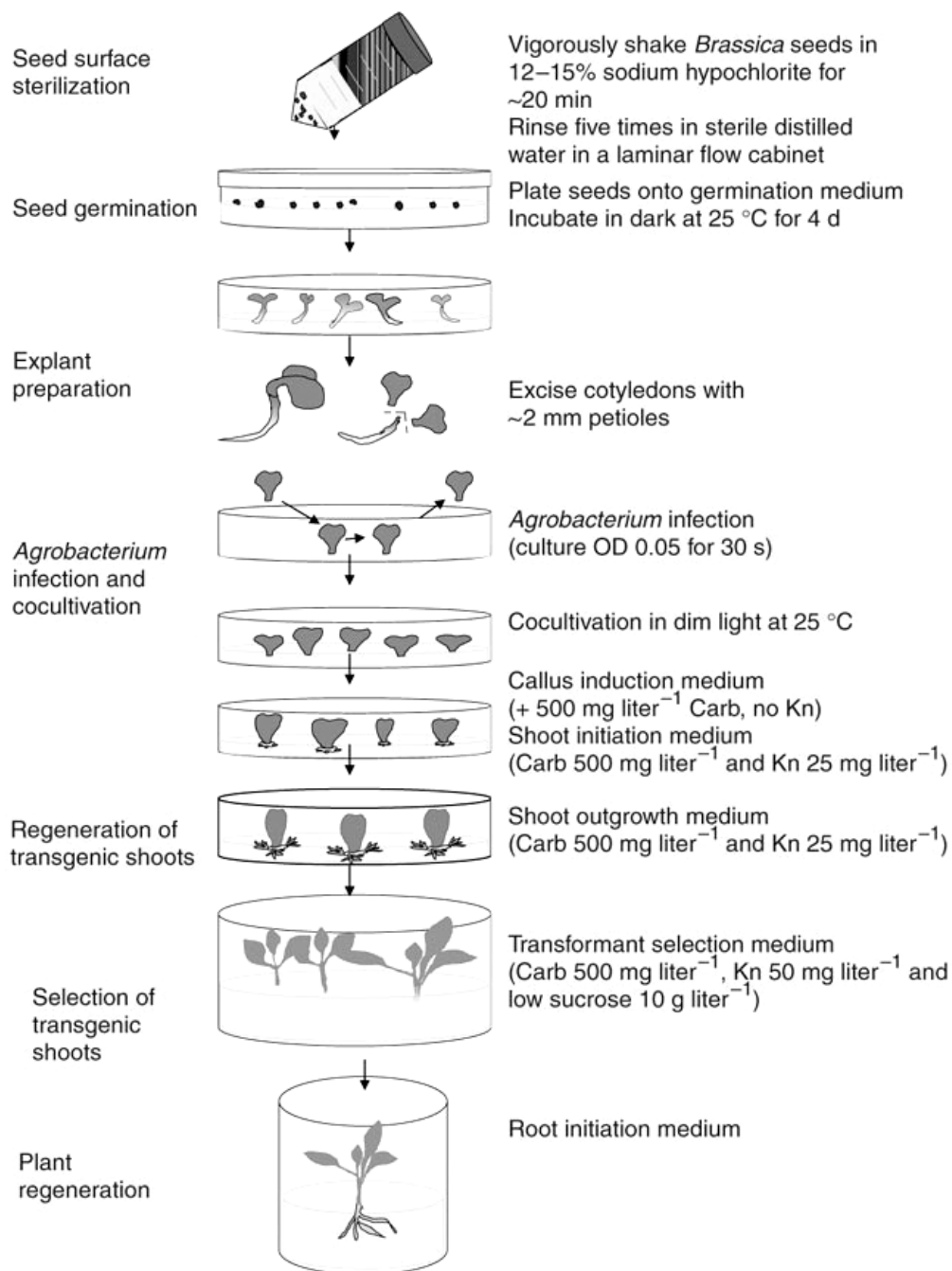


Fig. 1.1

Legend: Carb = carbenicillin; Kn = kanamycin

- (d) With reference to Fig. 1.1, explain why carbenicillin and kanamycin were added in the various media.

.....
.....
.....[3]

- (e) Explain why plant tissue culture technique is used to regenerate the transformed plants.

.....
.....[2]

- (f) State the difference in composition between the root initiation medium and the shoot outgrowth medium.

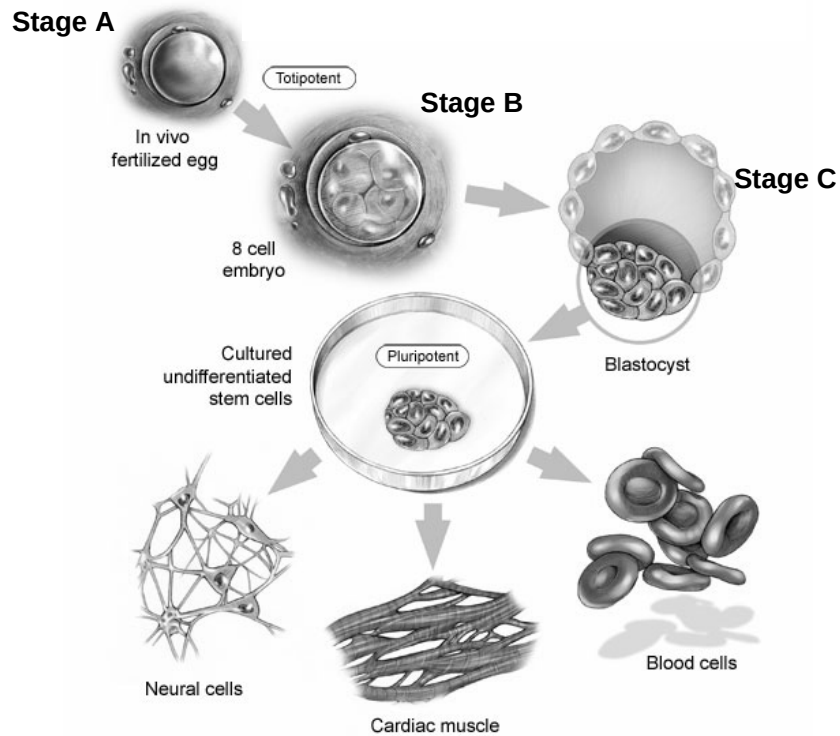
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.....[1]

- (g) Suggest why seeds were used instead of explants from adult *Brassica* plants, as starting plant material for this procedure.

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.....[1]

[Total: 14]

- 2 Fig. 2.1 shows the development stages of cells following a fertilization event.



Source: <http://www.csa.com/discoveryguides/stemcell/overview.php>

Fig. 2.1

- (a) Identify the stage(s) in Fig. 2.1 where zygotic stem cells are found.

.....
[1]

- (b) Distinguish between zygotic stem cells and adult stem cells.

.....

[3]

The Ministry of Health has currently approved a number of cases in the use of umbilical cord blood in the treatment of Cerebral Palsy. Cerebral Palsy is characterised by damage to the part of the brain that controls motor functions. As such, the child suffering from this condition faces challenges in movement, learning, hearing, seeing, and thinking. It has been reported that the patients generally experience an improvement in the condition but require repeated treatments.

- (c) (i)** Explain why using umbilical cord blood will not yield the best results for Cerebral Palsy.

.....

[1]

- (ii)** State and explain two other types of stem cells which will be more appropriate for use as treatment.

.....

[4]

The stem cells can be donated and stored in a cord blood bank. Some examples of the stored stem cells are bone marrow, peripheral blood and umbilical cord blood. All these stem cells are used in transplant procedures.

- (d)** State the advantages and limitations of using umbilical cord blood stem cells as compared to bone marrow and peripheral blood stem cells.

.....

[4]

[Total: 13]

[Turn over

- 3 Human DNA polymerase is involved in DNA replication and repair. The enzyme exists as a complex which comprises four subunits and more than 1 binding site. During DNA replication the enzyme synthesizes a new strand of DNA by extending an existing nucleotide chain. Fig. 3.1 shows part of the structure of a human DNA polymerase enzyme associating with DNA.

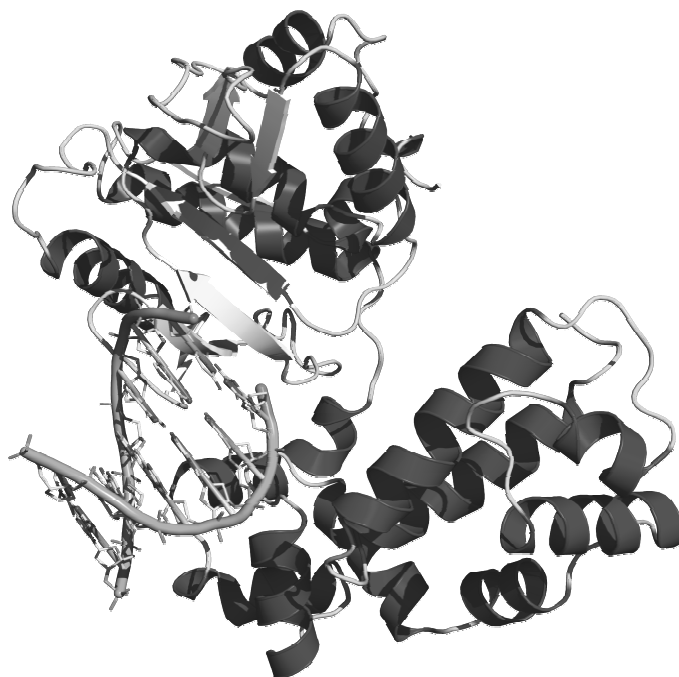


Fig. 3.1

- (a) (i) With reference to the information given and Fig. 3.1, describe the structure of human DNA polymerase.

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.....[4]

- (ii) Explain how the structure of human DNA polymerase enables it to carry out its role in DNA replication.

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.....[2]

In 1965, Thomas D. Brock isolated the enzyme *Taq* DNA polymerase from the bacterium *Thermus aquaticus*. This enzyme is frequently used in the polymerase chain reaction (PCR), a technique used to replicate short segments of DNA.

(b) Compare DNA replication by *Taq* DNA polymerase and human DNA polymerase.

.....

[2]

The sequence of nucleotides in a sample of DNA can be found using a method involving PCR followed by gel electrophoresis. In this method, four sequencing reactions are carried out in separate tubes at the same time.

Each tube contains:

- DNA polymerase
- radioactively-labelled primers
- template DNA which is single-stranded
- a mixture of DNA nucleotides (dTTP, dATP, dGTP, dCTP)
- one of four types of modified nucleotides (ddTTP, ddATP, ddGTP, ddCTP)

The modified nucleotide is also known as a replication-stopping nucleotide. The incorporation of a modified nucleotide during the synthesis of the new strand stops the replication process.

Fig. 3.2 shows the copies produced by the PCR from a strand of DNA using a mixture of DNA nucleotides where some of the thymine-containing nucleotides were modified.

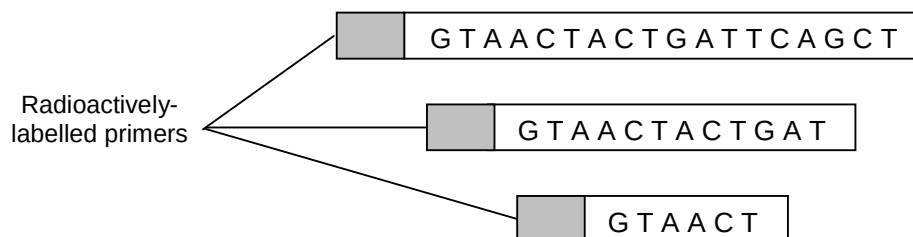


Fig. 3.2

(c) Explain the use of radioactively-labelled primers in this method of DNA sequencing.

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[2]

(d) With reference to Fig. 3.2,

(i) deduce the sequence of the template DNA strand.

.....[1]

(ii) state the sequence of the shortest DNA fragment that would be produced in the tube containing modified adenine.

.....[1]

Fig. 3.3 shows the positions of the bands produced from the tubes containing ddTTP and ddATP.

(iii) On Fig. 3.3, draw the bands you would observe from the tubes containing ddGTP and ddCTP. [1]

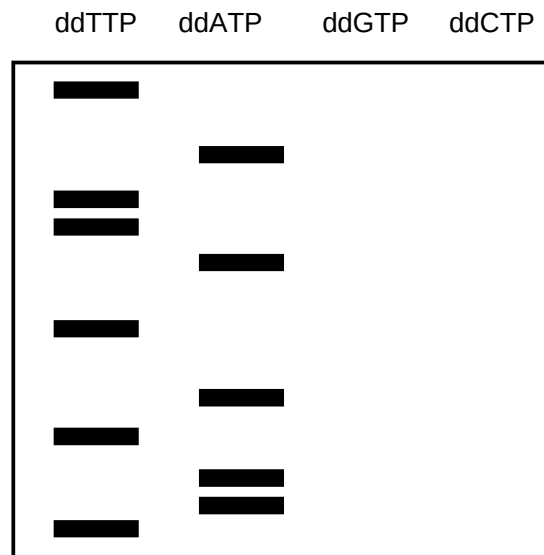


Fig. 3.3

[Total: 13]

Planning Question

- 4 You are required to plan, but not carry out an investigation into the effect of increasing the concentration of glucose on the respiration rate of yeast.

Yeast, in the presence of glucose, undergoes respiration. Yeast enzymes catalyse the breakdown of glucose to carbon dioxide and water.

You are to use methylene blue, which acts as an artificial electron acceptor during respiration. Methylene blue changes from blue to colorless as a result of its reduction in the enzyme reaction.

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- Yeast suspension
- 10% glucose solution
- Distilled water
- Access to tap water and a water bath maintained at 65°C
- Methylene blue
- Thermometer
- Stopwatch

[Total: 12]

Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labeled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

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[Turn over

[illegible]

This image shows a full page of white paper designed for handwriting practice. It features approximately 20 evenly spaced horizontal dotted lines running across the width of the page. There are no margins, text, or other markings present.

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 (a) Describe the uses of a cDNA library and the steps in creating one. [4]

(b) Describe how gene therapy can be used to treat cystic fibrosis. [9]

(c) Outline how Restriction Fragment Length Polymorphisms can be used for genomic mapping and disease detection. [7]

[Total: 20]